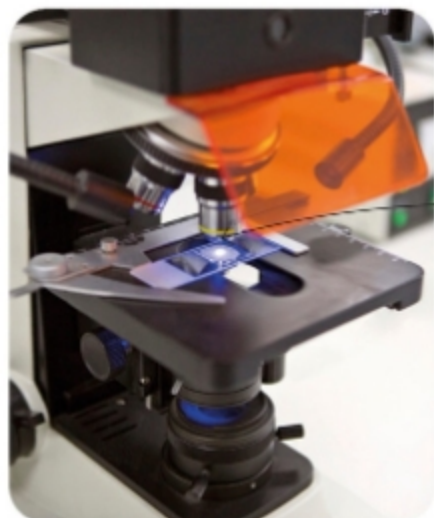


Cadmium

48
Cd

This element was discovered in 1817 in a zinc-based mineral called calamine. In its pure form, the soft metal is silvery with a bluish tinge. Once a common ingredient in a variety of products, such as paint, cadmium is now known to be toxic and is used less and less.



Ultraviolet (UV) light in this microscope is produced by a cadmium laser.

This brittle mineral is easily chipped.



Pure cadmium pellet refined in a lab

Smithsonite

Named after the English chemist James Smithson, this mineral was first identified in 1802. Smithsonite mainly contains white zinc carbonate, but cadmium impurities give it a bright yellow color.

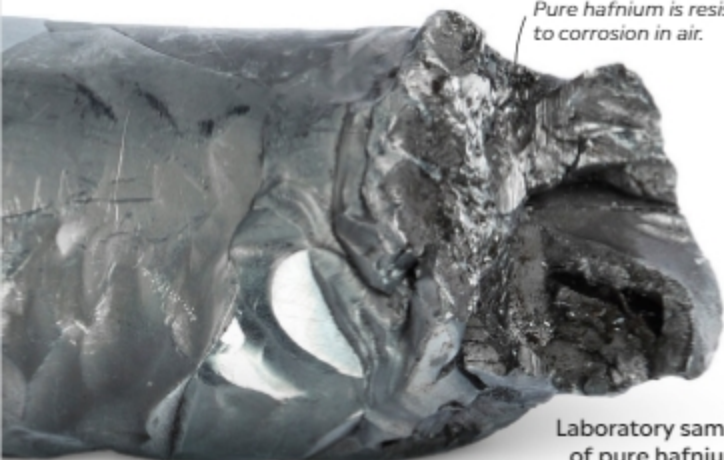
High-intensity lasers

Powerful optical microscopes use cadmium lasers to study tiny specimens, such as microscopic organisms. The data collected by these lasers can then be pieced together to create 3-D images of the specimens, so scientists can study them further.

Hafnium

72
Hf

Dmitri Mendeleev's prediction in 1869 about the existence of "element 72" encouraged Dutch physicist Dirk Coster and Hungarian radiologist Georg Karl von Hevesy to study zirconium ores. They discovered hafnium in 1923. Today, hafnium compounds are used to make specialized electronic circuits.



Pure hafnium is resistant to corrosion in air.

Laboratory sample of pure hafnium



Rods of pure tantalum refined in a laboratory

Tantalum shell is flexible and lightweight.



Walking easy

Tantalum body implants are more porous than titanium, and tantalum bonds directly to bone, which encourages bone growth and keeps the implant firmly in place. Its ability to mimic the elasticity of bones has made tantalum an excellent choice for patients who need an artificial joint.

Tantalum

Artificial hip implant

73
Ta

A hard metal that does not corrode easily, tantalum is combined with softer metals to make them stronger. Capacitors—devices used to store electric charge—in gadgets such as cell phones are often made of tantalum powder. It is also used to make artificial joints.